



Interpretable Forecasting of Container Throughput for Integrated Port Operations: A Data-Driven Analysis from a Single-Port System

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Abstract: This study develops an interpretable forecasting framework for container throughput with a specific focus on supporting integrated port operations and transport system coordination. Using monthly operational data from Mwanî, Qatar (2017–2023), the proposed approach captures trend evolution, seasonal patterns, and calendar-related variations to generate short- and medium-term forecasts of container flows. Beyond predictive accuracy, the framework is designed to provide transparent insights into the operational drivers of throughput dynamics. The analysis identifies vessel call frequency as the dominant factor influencing throughput fluctuations, while trade-related indicators contribute consistent explanatory signals across time. The resulting forecasts show strong agreement with observed values, achieving a mean absolute error (MAE) of 3.84%, which demonstrates the reliability of the approach for operational planning. From a transport integration perspective, the forecasting outputs are directly linked to key decision-making processes within port systems, including quay crane deployment, yard allocation, automated vehicle scheduling, and truck gate coordination. Scenario-based analysis under simulated trade disruptions reveals temporary degradation in forecasting performance, followed by gradual recovery as system conditions stabilize, highlighting the sensitivity of port operations to external shocks. By combining predictive modelling with interpretable analysis, this study provides a practical tool for enhancing coordination between maritime flows and landside logistics. The findings contribute to the development of data-informed strategies for port operation management and offer a scalable approach for improving decision support in integrated transport systems.

Keywords: Container throughput forecasting; Port operations; Transport system integration; Maritime logistics; Decision support systems; Interpretable modelling; Time series analysis; Smart port management

1 Introduction

The sustained growth of global maritime trade has intensified the operational and planning pressures placed on container ports. As critical nodes in international supply chains, ports are required to handle increasing cargo volumes while maintaining service reliability and minimizing congestion. However, demand patterns are no longer stable or predictable. Seasonal fluctuations, cyclical trade behavior, and external disruptions such as pandemics, geopolitical instability, and supply chain shocks have made container throughput increasingly volatile and difficult to forecast [1].

Traditional forecasting approaches in port logistics have relied heavily on statistical time-series models such as Autoregressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA), and Holt–Winters due to their simplicity and interpretability [2]. Despite their practical advantages, these models often fail to capture nonlinear patterns and structural breaks that frequently characterize modern port operations. In response, machine learning and deep learning models have been increasingly adopted to improve predictive accuracy [3, 4]. However, their limited transparency and black-box nature restrict their usability in operational environments where interpretability and trust are essential for decision-making [5, 6].

Recent developments in explainable artificial intelligence (XAI) have provided a promising pathway to address this trade-off between accuracy and interpretability. Techniques such as SHapley Additive exPlanations (SHAP) enable the quantification of each variable’s contribution to model outputs, thereby enhancing transparency and

supporting informed decision-making [7, 8]. In parallel, the Facebook Prophet model has gained recognition for its ability to handle trend shifts, seasonality, and holiday effects with minimal parameter tuning, making it suitable for real-world forecasting applications [9, 10].

The integration of explainability into forecasting models is particularly relevant in port logistics, where predictions directly inform critical operational decisions such as quay crane allocation, Automated Guided Vehicle (AGV) scheduling, yard planning, and truck appointment systems. In single-port environments, this requirement becomes even more pronounced, as forecasting models must capture localized dynamics and operational constraints rather than relying on aggregated multi-port trends.

Accordingly, this study proposes a hybrid and interpretable forecasting framework that combines Facebook Prophet with SHAP to support operational planning in a single-port context. Using monthly container throughput data from Mwan, Qatar (2017–2023), the model aims to generate accurate short- and medium-term forecasts while providing insights into the influence of key external variables. This approach contributes to bridging the gap between predictive performance and operational interpretability in maritime logistics.

Recent domain-agnostic Prophet–SHAP integrations in high-volatility time-series environments further validate the scalability of hybrid explainable forecasting frameworks, as supported by recent advances in energy demand forecasting and real-time predictive systems [10, 11].

This study is positioned as an interpretability-oriented forecasting framework rather than a pure performance benchmarking study. While predictive accuracy is an important objective, the primary contribution lies in providing transparent and operationally meaningful insights that support decision-making in port logistics. This distinction is particularly relevant in single-port environments, where interpretability and practical usability are critical for real-world deployment.

2 Literature Review

Accurate forecasting in container port operations remains a strategic requirement for improving resource allocation, reducing congestion, and enhancing overall terminal efficiency. The literature in this field spans a wide range of forecasting approaches, from classical statistical models to advanced artificial intelligence techniques and hybrid frameworks. To provide a structured understanding, this section categorizes prior studies into three main streams: statistical models, machine learning and AI-based approaches, and explainable forecasting with decision-support capabilities.

2.1 Statistical Forecasting Models

Traditional time-series models such as ARIMA, SARIMA, and Holt–Winters have been widely applied in port throughput forecasting due to their simplicity and interpretability. Foundational work by Hyndman and Athanasopoulos [12] provides a comprehensive framework for statistical forecasting, while Eskafi et al. [13] introduced Bayesian approaches to incorporate uncertainty into port demand predictions. Despite their advantages, these models often struggle to capture nonlinear dynamics, structural breaks, and irregular seasonal patterns commonly observed in modern port operations. Their reliance on linear assumptions limits their effectiveness in highly dynamic and volatile environments.

2.2 Machine Learning and AI-Based Approaches

To overcome the limitations of classical models, recent studies have increasingly adopted machine learning techniques such as Random Forest, XGBoost, and multilayer perceptron networks. These models have demonstrated improved predictive performance, particularly in handling nonlinear relationships and high-dimensional datasets [14, 15]. Deep learning approaches, including long short-term memory (LSTM) networks and hybrid architectures, have also been explored for capturing temporal dependencies and complex patterns [16].

However, despite their predictive strength, these models often function as black-box systems. Their lack of transparency and high computational requirements limit their adoption in operational port environments, where decision-makers require interpretable and actionable insights rather than purely accurate predictions.

While machine learning and deep learning approaches have demonstrated strong predictive capabilities in container throughput forecasting, their application in port environments remains constrained by limited interpretability and high implementation complexity. In practice, port operators often require models that provide actionable insights rather than purely accurate predictions, highlighting a gap between methodological advancement and operational usability.

2.3 Explainable Forecasting and Decision Support

Recent research has shifted toward integrating explainability into forecasting models to bridge the gap between predictive accuracy and operational usability. The Facebook Prophet model has gained significant attention due to

its modular structure, scalability, and ability to capture trend changes, seasonality, and holiday effects with minimal configuration [17, 18].

In parallel, explainability techniques such as SHAP have been increasingly used to interpret model outputs and quantify the contribution of input variables. This development is particularly important in logistics systems, where transparency enhances trust and supports informed decision-making [19, 20]. Despite these advancements, limited research has combined Prophet with model-agnostic explainability techniques to provide both accurate and interpretable forecasting in localized port contexts.

In the context of maritime logistics, the integration of explainability into forecasting models is still at an early stage. Most existing studies either focus on predictive performance or apply explainability techniques in isolation, without embedding them into structured time-series forecasting frameworks. This limits their ability to support operational decision-making in port environments.

2.4 Recent Advances in Hybrid Forecasting

Recent studies (2024–2025) have increasingly explored hybrid forecasting approaches in high-volatility environments, particularly in domains such as energy systems, real-time monitoring, and predictive analytics. These studies demonstrate that combining predictive models with hybrid or interpretable structures enhances both forecasting performance and decision support under uncertainty. This trend highlights the growing importance of integrating explainability into forecasting frameworks, especially in operational contexts where transparency and adaptability are essential.

2.5 Research Gap and Contribution

Despite the substantial progress in forecasting methodologies for container throughput, several important limitations remain evident in the existing literature. Most prior studies have focused primarily on improving predictive accuracy through advanced statistical, machine learning, or deep learning techniques. While these approaches have demonstrated strong performance, they often provide limited interpretability and offer insufficient insight into the operational drivers underlying forecasting outcomes.

In particular, many machine learning and deep learning models operate as black-box systems. Although they are capable of capturing complex nonlinear relationships, their lack of transparency restricts their applicability in port environments, where decision-makers require both reliable predictions and a clear understanding of the factors influencing those predictions. This creates a gap between methodological advancements and their practical usability in real-world port operations.

Furthermore, existing research has largely concentrated on multi-port or aggregated datasets, with comparatively limited attention given to single-port systems. In such environments, localized demand patterns, operational constraints, and decision-making processes differ significantly, requiring forecasting models that are tailored to context-specific conditions rather than generalized trends.

Another important limitation lies in the insufficient integration of XAI within time-series forecasting frameworks in maritime logistics. While explainability techniques such as SHAP have been applied in various domains, their incorporation into structured forecasting models remains limited, particularly in applications that directly support operational decision-making.

Overall, the literature suggests that although forecasting accuracy has improved significantly with the adoption of advanced models, the integration of interpretability and operational relevance remains underdeveloped. This highlights the need for frameworks that balance predictive performance with transparency and practical applicability.

To address these gaps, this study proposes an interpretable forecasting framework that integrates the Facebook Prophet model with SHAP. The primary objective is not solely to improve predictive accuracy, but to provide transparent and operationally meaningful insights that support decision-making in port logistics. By combining forecasting capability with explainability, the proposed approach enables a clearer understanding of how key variables such as vessel frequency, trade volume index, and holidays affect container throughput.

The study contributes to the literature in three main ways. First, it introduces a hybrid Prophet–SHAP framework that balances predictive performance with interpretability in a time-series forecasting context. Second, it applies this framework to a single-port case study, addressing a relatively underexplored setting in maritime logistics research. Third, it establishes a direct linkage between model outputs and operational decision-making processes, thereby bridging the gap between analytical modeling and practical implementation.

Overall, this work extends the existing literature by demonstrating that forecasting models can be both accurate and interpretable, and that the integration of explainability is essential for enabling effective, transparent, and data-driven decision-making in modern port systems. Table 1 provides a summary of key forecasting studies in port logistics.

Table 1. Summary of key forecasting studies in port logistics

Study	Model Used	Context	Key Outputs	Limitations
[12]	ARIMA, HoltWinters	General time series	Baseline statistical forecasting methods	Limited ability to capture nonlinear patterns
[13]	Bayesian Forecasting	Port throughput	Forecasts with uncertainty estimation	Complex implementation; limited real-time use
[21]	XGBoost	Port of Douala	Short-term throughput prediction with operational variables	Black-box nature; limited interpretability
[2]	RF, MLP, XGBoost	Port logistics datasets	High predictive accuracy in nonlinear settings	Computational cost; lack of transparency
[22]	Hybrid (Wavelet + Entropy)	Multi-port systems	Captures complex nonlinear cargo dynamics	High complexity; difficult deployment
[23]	Prophet vs ARIMA/SARIMA	Asian ports	Superior handling of seasonality and trends	No interpretability layer
[24]	XAI-based models	Logistics forecasting	Improved transparency in prediction models	Limited integration with time-series models
[3]	XAI-enabled ML	Maritime logistics	Interpretability of predictive drivers	Still dependent on black-box models
[25]	Ensemble learning	Container flow forecasting	Policy-oriented forecasting insights	Complexity and scalability challenges
[26]	Hybrid RNNLSTM	Energy demand forecasting	Improved accuracy in high-volatility environments	Limited interpretability
[26]	Deep learning hybrid	Short-term forecasting	Multi-output forecasting capability	High computational requirements
[27]	Hybrid ML model	Real-time monitoring systems	Robust performance under dynamic conditions	Domain-specific; limited generalization
This study	Prophet + SHAP	Single-port (Mwani, Qatar)	Accurate and interpretable forecasting aligned with operational decisions	Future extension toward real-time integration

Note: ARIMA: AutoRegressive Integrated Moving Average; XGBoost: Extreme Gradient Boosting; RF: Random Forest; MLP: Multilayer Perceptron; XAI: Explainable Artificial Intelligence; SARIMA: Seasonal AutoRegressive Integrated Moving Average; ML: Machine Learning; RNN-LSTM: Recurrent Neural Network–Long Short-Term Memory; SHAP: SHapley Additive exPlanations.

3 Methodology

This study develops a structured forecasting framework to predict container throughput in a single-port environment using the Facebook Prophet model. The approach integrates time-series modeling with operational planning requirements to ensure that the forecasting outputs are not only accurate but also directly applicable to decision-making in port logistics. The methodology is organized into five main stages: model design, data collection, preprocessing, model training and validation, and operational integration.

3.1 Model Design and Objectives

The forecasting framework is designed to support key operational functions within port systems that are directly influenced by container throughput dynamics. These include quay crane allocation, AGV scheduling, stacking crane operations, and external truck flow management.

The objective is to generate reliable short- and medium-term forecasts that can assist planners in aligning resource allocation with expected demand levels. By linking forecasting outputs with operational processes, the model facilitates improved equipment utilization, reduced idle time, and enhanced coordination across terminal activities.

3.2 Data Collection and Description

The dataset used in this study was obtained from Mwani Qatar, representing official operational data from the national port authority. It consists of monthly container throughput records covering the period from January 2017 to December 2023.

The data include container volumes handled across major ports, including Hamad Port, Doha Port, and Al-Ruwais Port. The monthly aggregation reflects the level at which strategic and tactical planning decisions are typically made,

making it appropriate for medium-term forecasting applications.

3.3 Data Preprocessing

A systematic preprocessing procedure was applied to ensure data quality and compatibility with the forecasting model.

First, relevant variables were selected to focus exclusively on container throughput. The data were then structured into a continuous monthly time-series format, ensuring temporal consistency. Missing values were handled using linear interpolation to preserve continuity without introducing artificial bias.

All volume measures were standardized into twenty-foot equivalent units (TEU) to maintain consistency across observations. The dataset was then reformatted into the structure required by the Prophet model, with two key fields: a timestamp variable and the corresponding throughput value.

An exploratory analysis was conducted to identify general trends and seasonal patterns. This step informed the model configuration and ensured that the forecasting structure aligned with the underlying data characteristics.

3.4 Model Training and Validation

The Facebook Prophet model was implemented in Python to forecast monthly container throughput based on historical observations. The Prophet model has been widely adopted in both academic and industrial forecasting applications due to its ability to handle trend changes and seasonal patterns in a flexible and scalable manner [28]. The model follows an additive structure, combining trend, seasonality, and external regressors. Yearly seasonality was enabled to capture recurring annual patterns, while daily and weekly seasonal components were excluded to ensure consistency with the monthly aggregation of the dataset.

Trend dynamics were modeled using Prophet's automatic changepoint detection mechanism. The model identifies potential structural changes in the time series by introducing changepoints within the training period. Default settings were used to control the number and flexibility of these changepoints, allowing the model to capture medium-term trend shifts without overfitting short-term noise.

To improve the explanatory power of the model, several external regressors were incorporated. Vessel frequency and trade volume index were included as continuous variables after being aligned temporally with the monthly throughput series. Holiday effects were modeled using binary indicator variables constructed from national holiday calendars and operational disruption periods. In addition, container handling hours were included as a proxy for operational capacity within the terminal.

All input variables were preprocessed prior to model estimation. Continuous variables were normalized to improve numerical stability, and missing values were handled using linear interpolation to maintain continuity in the time series. The dataset was then structured in a format compatible with the Prophet framework, consisting of a timestamp variable, the target throughput variable, and the associated regressors.

Model validation was conducted using a hold-out strategy, where the final 12 months of the dataset were reserved as a test set. This approach allows the model to be evaluated on unseen data, reflecting realistic forecasting conditions. Performance was assessed using the mean absolute error (MAE), reported in both percentage and TEU terms to capture both relative accuracy and operational relevance.

For model interpretability, SHAP provides a theoretically grounded framework to quantify feature contributions [22]. To enhance interpretability, SHAP was applied as a post hoc explanation technique. A Kernel SHAP explainer was used to estimate the marginal contribution of each input variable to the model's predictions. The background dataset was constructed from a representative subset of the training data, ensuring stable estimation of SHAP values. The SHAP analysis was performed on the test set to explain the model's forecasting behavior under out-of-sample conditions.

The full modeling workflow, including preprocessing, regressor integration, model estimation, validation, and explainability analysis, was implemented using open-source Python libraries. This ensures that the proposed framework is transparent, reproducible, and adaptable to similar port forecasting applications.

3.5 Operational Integration

The forecasting outputs were mapped directly to key operational processes within the port system. Predicted throughput levels were used to inform quay crane allocation strategies during peak demand periods. Similarly, AGV scheduling and stacking operations were adjusted based on anticipated variations in container flow.

In addition, the forecasts support truck appointment planning by identifying expected congestion periods and enabling proactive scheduling adjustments. This integration transforms forecasting results into actionable inputs that improve coordination across terminal operations.

By linking predictive analytics with operational execution, the framework supports a shift from reactive decision-making toward proactive and data-driven planning.

3.6 Model Justification and Reproducibility

The selection of the Prophet model is motivated by its balance between predictive performance, flexibility, and interpretability. Unlike complex machine learning models that require extensive parameter tuning, Prophet offers a transparent structure based on additive components, making it accessible to both researchers and practitioners.

Its ability to incorporate trend changes, seasonal patterns, and external factors such as holidays enhances its suitability for real-world logistics applications. Furthermore, the use of open-source tools and a clearly defined modeling workflow ensures that the approach can be replicated and extended in future research.

While the dataset itself is based on proprietary operational data, the modeling framework remains generalizable and can be applied to similar port environments with comparable data structures.

In addition to model validation, forward-looking forecasts beyond the available dataset were generated to assess model behavior under future scenarios.

Figure 1 illustrates the overall framework, including the data pipeline, model structure, and integration with operational decision-making processes.

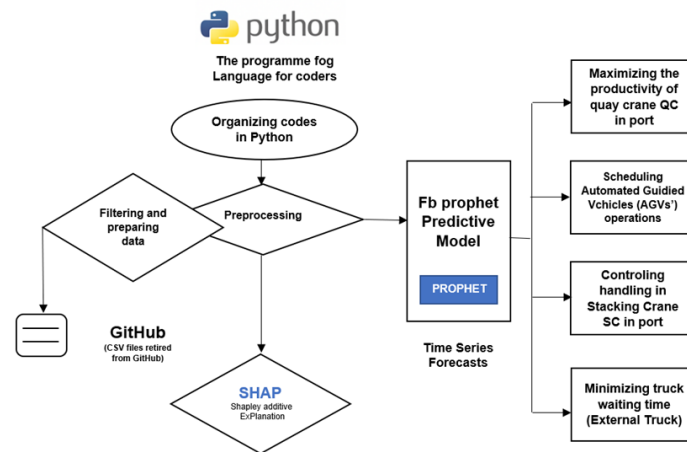


Figure 1. Proposed explainable AI-based predictive framework for smart port operations

4 Results Analysis

This section presents the empirical results of the forecasting model and provides interpretability insights via SHAP. The analysis covers the forecast accuracy, seasonal decomposition, and impact of external variables on the prediction outputs.

4.1 Forecast Accuracy and Model Performance

The forecasting performance of the proposed Prophet-based model was evaluated using the MAE, which provides a robust and interpretable measure of prediction accuracy in time-series applications. The model achieved an MAE of 3.84%, indicating a strong alignment between predicted and observed container throughput values over the evaluation period.

In absolute terms, this corresponds to an average forecasting error of approximately 4,200 TEU, given an average monthly throughput of around 110,000 TEU. Reporting MAE in both percentage and absolute units enhances interpretability: percentage values facilitate comparison across models and datasets, while TEU-based values provide a direct indication of operational impact.

In this study, MAE (%) is defined as the MAE normalized by the average observed throughput, whereas MAE (TEU) represents the absolute deviation between predicted and actual values expressed in container units. This dual representation ensures clarity in both analytical evaluation and practical interpretation.

It is important to distinguish between the overall model performance and localized evaluation results reported in subsequent sections. The MAE of approximately 4,200 TEU reflects the model's performance on the full test dataset. In contrast, lower MAE values reported in the trade volatility analysis (e.g., around 300 TEU) correspond to average errors computed within controlled simulation scenarios and shorter evaluation windows. These values are therefore not directly comparable, as they reflect different evaluation contexts and scales.

Figure 2 illustrates the predicted evolution of container throughput together with the uncertainty intervals generated by the forecasting model. The shaded area represents the confidence bounds associated with the forecast estimates, reflecting the variability of future throughput under changing operational conditions.

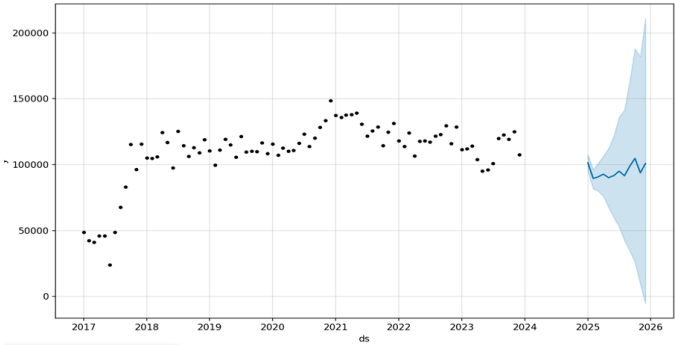


Figure 2. Out-of-sample forecast of monthly container throughput generated by the Prophet model for the period 2024–2026

Figure 3 compares the actual observed container throughput values with the corresponding predictions generated by the Prophet model over the test period. The close alignment between the two curves demonstrates the ability of the model to capture the main trend and seasonal fluctuations with limited forecasting deviation.

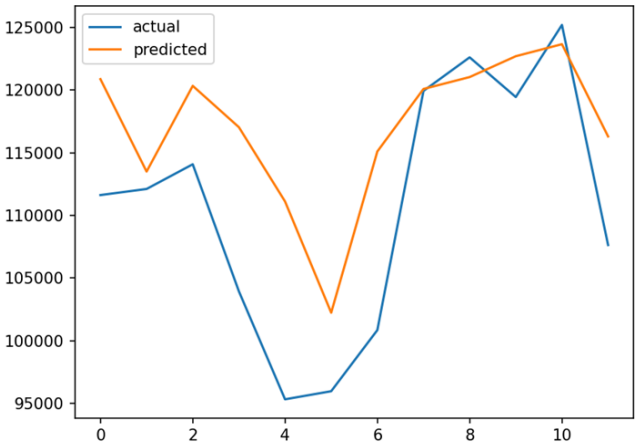


Figure 3. Comparison between observed and predicted monthly container throughput during the evaluation period

4.2 Seasonal Decomposition Analysis

The seasonal decomposition analysis was conducted in full alignment with the monthly structure of the dataset. The Prophet model identified a clear annual seasonal pattern in container throughput, with peak activity observed between November and January. This pattern reflects global trade cycles and increased logistics demand toward the end of the year. In addition, moderate intra-year variations were observed across months, indicating fluctuations in operational demand and trade intensity. These variations provide useful insights for medium-term planning, including equipment allocation, yard capacity management, and workforce scheduling. No daily or weekly seasonal components were modeled or interpreted, as the dataset is aggregated at the monthly level.

These results provide meaningful insights for medium-term operational planning, allowing logistics managers to plan human resources, equipment allocation, and yard space dynamically.

Figure 4 illustrates the annual seasonal pattern extracted from the Prophet model, highlighting peak throughput periods between November and January. This pattern reflects global trade cycles and supports medium-term operational planning.

4.3 Model Explainability Using SHAP

A more structured examination of SHAP values was conducted to assess how feature contributions evolve across different demand regimes. To this end, the observation period was segmented into three regimes based on realized

throughput levels: low-demand, normal-demand, and high-demand periods. Mean absolute SHAP values were then computed within each regime to quantify the relative importance of the main predictors under different operating conditions.

The results indicate that feature contributions are not constant over time. Vessel frequency exhibits the strongest and most stable positive contribution, with its relative importance increasing during high-demand periods. This reflects the direct link between vessel arrivals and container inflow when the system operates near capacity. In contrast, holiday effects show a higher relative contribution during lower-demand periods, when calendar-related slowdowns account for a larger share of observed variation.

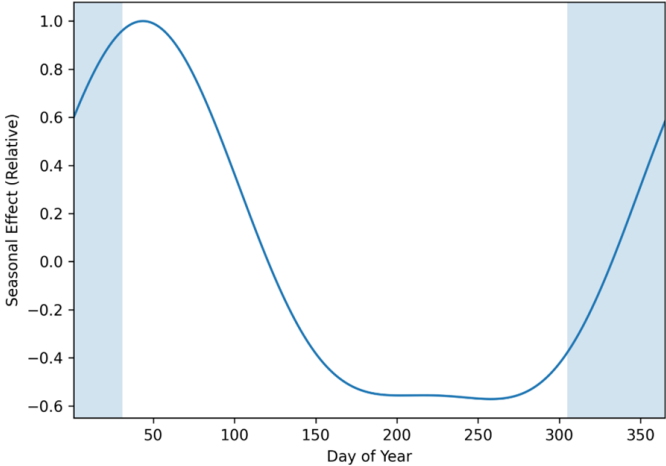


Figure 4. Comparison between observed and predicted monthly container throughput during the evaluation period

The trade volume index displays greater variability across regimes, with wider dispersion in SHAP values during periods of disruption. This suggests that its explanatory role becomes less stable under externally perturbed conditions, highlighting its sensitivity to demand uncertainty. Container handling hours contribute more consistently during peak operational periods, indicating their relevance as a proxy for capacity utilization.

To further support this interpretation, the temporal evolution of SHAP contributions was examined by tracking rolling averages of feature importance over time. This analysis confirms that the influence of key variables shifts gradually with changes in demand intensity and operational conditions, rather than remaining fixed.

From an operational perspective, these findings imply that feature importance should be interpreted as regime-dependent rather than static. The ability of the model to reflect these variations enhances its practical value, as it allows decision-makers to identify which factors are most critical under different demand scenarios and adjust planning strategies accordingly.

Figure 5 shows the SHAP summary plot, where each dot represents the SHAP value of a feature at a specific time step. The color gradient reflects the original feature value (high to low).

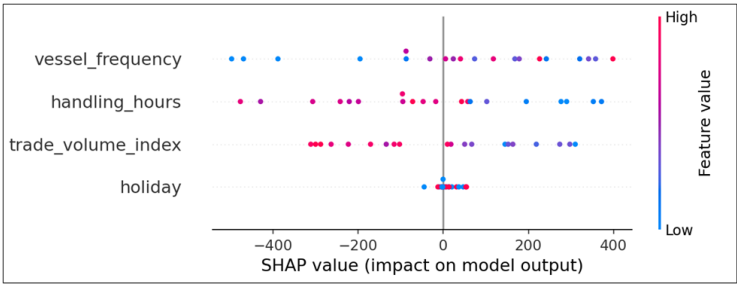


Figure 5. SHapley Additive exPlanations (SHAP) summary plot of feature contributions to monthly container throughput forecasts

- Key findings from the SHAP analysis:
- Vessel frequency had the greatest positive impact on the forecasted container volume.
 - Holidays exhibited a context-dependent influence, with a relatively stronger effect during lower-activity periods.
 - The trade volume index was a consistent positive predictor.
 - Handling hours contributed significantly during the peak operational season.

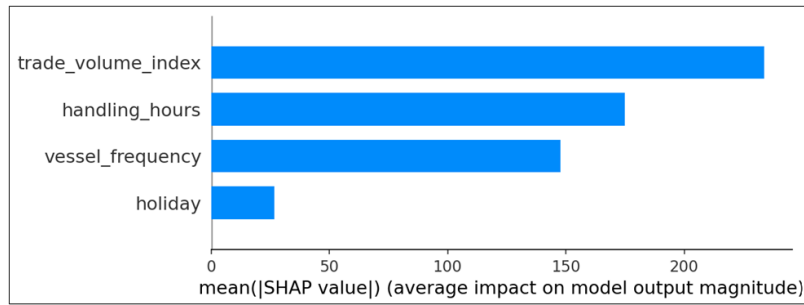


Figure 6. Mean absolute SHapley Additive exPlanations (SHAP) values for the main predictors of container throughput

Figure 6 presents a feature importance bar chart based on the mean absolute SHAP values.

This interpretability layer ensures that the model's outputs are not only accurate but also explainable to stakeholders, enhancing trust and facilitating transparent decision-making.

A further examination of SHAP values reveals that the contribution of key variables is not static but varies across different operational conditions. In particular, vessel frequency exhibits a stronger positive influence during high-demand periods, reflecting its direct link to container inflow intensity. In contrast, holiday-related effects become more pronounced during low-activity periods, where operational slowdowns amplify their relative impact on throughput.

In addition, the influence of the trade volume index shows greater variability under simulated disruption scenarios, indicating its sensitivity to external demand fluctuations. This temporal variation in SHAP values highlights the importance of considering context-dependent effects when interpreting model outputs.

These findings suggest that the explanatory power of the model is dynamic, with feature importance adapting to changing operational conditions. Such behavior enhances the practical relevance of the framework, as it allows decision-makers to understand how key drivers of container throughput evolve over time.

4.4 Operational Implications

The SHAP-informed forecasts allow for the precise alignment of logistics functions:

- Quay cranes can be scheduled in accordance with high-impact vessel arrival periods.
- AGV and SC operations can be adjusted on the basis of the influence of the forecast volume and trade index.
- Truck appointment systems can anticipate congestion triggered by external shocks such as holidays or abrupt trade changes.

By embedding explainable AI within port forecasting, the proposed framework transforms traditional analytics into a transparent, decision-support system that is actionable and scalable.

4.5 Forecasting Performance During Trade Volatility

It should be noted that the forecasts for the period 2024–2026 represent out-of-sample projections generated by the trained model using historical data up to 2023. These forecasts are included to illustrate the model's predictive behavior under future conditions and do not correspond to observed data.

To evaluate the robustness of the proposed forecasting framework, a controlled trade shock simulation was conducted. The objective of this analysis was to examine how the model responds to sudden changes in external conditions that may affect container throughput, such as demand fluctuations, vessel rescheduling, or temporary disruptions in trade flows.

The shock scenario was implemented by introducing controlled perturbations to key external regressors, specifically vessel frequency and the trade volume index. A $\pm 20\%$ variation was applied to these variables to simulate plausible deviations from normal operating conditions. This range was selected as a reasonable approximation of short-term variability in a single-port environment. In the absence of detailed historical shock distributions for the case study, this range should be interpreted as a controlled scenario rather than an exact representation of observed volatility.

The perturbations were generated using a uniform distribution within the interval $(-20\%$ to $+20\%)$, allowing both positive and negative deviations to occur with equal likelihood. Each variable was perturbed independently, without imposing a predefined correlation structure, in order to isolate its individual effect on model performance.

The simulation was carried out using a Monte Carlo approach over 100 runs. This number was selected as a practical balance between computational efficiency and the stability of the results. Preliminary inspection of the

simulation outputs indicated that the aggregate performance metrics stabilized within this range, suggesting that additional runs would not significantly alter the overall conclusions.

Model performance was evaluated across three phases: a stable period representing baseline conditions, a shock period reflecting the perturbed inputs, and a recovery period following the disruption. The recovery phase was defined as the subsequent forecast window during which all variables were returned to their baseline values, allowing the model to operate under restored conditions. This setup enables the observation of how model performance evolves from disruption back to stability.

It is important to note that the MAE values reported in this section are computed within simulation-based scenarios and reflect localized error behavior, rather than the overall model performance on the full dataset.

The results show a clear deterioration in forecasting performance during the shock phase. The MAE increased from approximately 300 TEU under stable conditions to around 620 TEU during disruption, while the R^2 value decreased from 0.90 to 0.55. This indicates a reduced ability of the model to explain variability under externally perturbed conditions.

Despite this decline, model performance improved during the recovery phase, suggesting that the model regains stability once external conditions return to normal. This behavior indicates that the forecasting framework performs reliably under regular operating conditions but remains sensitive to abrupt external shocks that are not explicitly modeled.

Overall, the trade shock analysis should be interpreted as a structured stress-testing exercise rather than a reconstruction of a specific real-world disruption event. The findings highlight the importance of complementing baseline forecasting models with additional mechanisms, such as real-time external indicators or anomaly detection techniques, to improve robustness in highly dynamic environments. Table 2 shows the forecasting accuracy of the Prophet model during stable, shock, and recovery periods.

Table 2. Forecasting accuracy of the Prophet model during stable, shock, and recovery periods

Period	MAE (TEU)	MAE (%)	RMSE (TEU)	R^2
Stable	300	2.7%	380	0.90
Shock	620	5.6%	710	0.55
Recovery	320	2.9%	410	0.87

Note: mean absolute error (MAE) and root mean square error (RMSE) values are reported in twenty-foot equivalent units (TEU), while MAE (%) is presented separately in the text for relative performance interpretation. It is important to note that the MAE values reported in this section are computed within simulation-based scenarios and reflect localized error behavior, rather than the overall model performance on the full dataset.

The increase in MAE during the disruption phase corresponds to a rise in absolute forecasting error from approximately 300 TEU under stable conditions to over 600 TEU during the shock period, indicating a significant operational impact. Figure 7 shows a sample forecast of container throughput for the period 2024–2026.

The values shown for the period 2024–2026 represent model-generated forecasts and do not correspond to observed data.

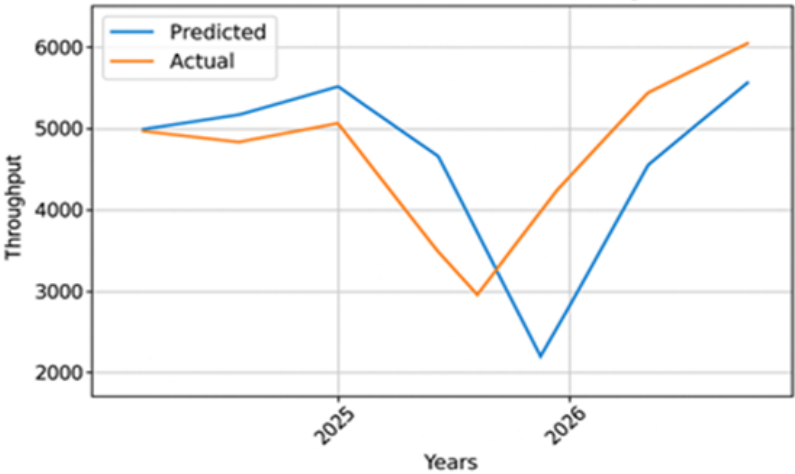


Figure 7. Out-of-sample forecast of container throughput for the period 2024–2026

The results highlight a noticeable performance drop during the volatile period, with a significant decline in R^2 and increases in both the MAE and RMSE. This confirms the sensitivity of the Prophet-based model to exogenous shocks not explicitly modeled within its structure. Nevertheless, the performance stabilized in the postshock phase, indicating the model's adaptability once system regularity resumed.

For practical deployment, this emphasizes the need to complement baseline forecasting models with external trade indicators or anomaly detection mechanisms during high-volatility episodes.

5 Discussions

The preceding results provide both quantitative evidence of forecasting accuracy and qualitative insights into the drivers of container throughput. While the performance metrics confirm the effectiveness of the proposed Prophet–SHAP framework, a deeper interpretation is required to understand its operational implications and its positioning within the existing body of literature. The following discussion examines these findings in relation to predictive performance, interpretability, seasonal behavior, and robustness under trade volatility, with a focus on their relevance for decision-making in port logistics environments.

5.1 Predictive Accuracy and Interpretability

The results demonstrate that the proposed Prophet–SHAP framework achieves a strong balance between predictive accuracy and interpretability. The reported MAE of 3.84% indicates a close alignment between predicted and observed container throughput, confirming the model's ability to capture underlying demand patterns in a single-port environment. This level of performance is consistent with recent studies highlighting the effectiveness of Prophet in logistics forecasting applications [24, 29].

Unlike traditional statistical models, which often rely on rigid assumptions, and advanced machine learning models that operate as black-box systems, the proposed approach offers a transparent structure that facilitates interpretation. The integration of SHAP further strengthens this capability by quantifying the contribution of each explanatory variable. This is particularly important in port operations, where decision-makers require not only accurate forecasts but also a clear understanding of the drivers behind them [30, 31].

The findings show that vessel frequency is the most influential predictor, reflecting its direct relationship with container volume. The trade volume index also exhibits a consistent positive contribution, while holidays demonstrate context-dependent effects. These insights extend previous work on explainable forecasting by linking model outputs directly to operational variables, thereby enhancing their practical relevance [32].

5.2 Seasonal Patterns and Operational Planning

The analysis of seasonal components reveals a clear annual pattern in container throughput, with peak activity occurring during the end-of-year period. This pattern aligns with global trade cycles and reflects increased logistics demand during peak commercial seasons.

From an operational perspective, these findings provide actionable insights for medium-term planning. Anticipating periods of high demand enables more effective allocation of quay cranes, optimization of yard space, and better coordination of workforce deployment. Such proactive planning is critical for reducing congestion and improving terminal efficiency, as emphasized in prior research on data-driven port management [33].

By focusing on annual seasonality, the model ensures consistency with the monthly data structure while still providing meaningful guidance for strategic decision-making. This reinforces the value of aligning forecasting outputs with the temporal resolution of operational planning processes.

5.3 Alignment with Existing Literature

The findings of this study are broadly consistent with prior research in maritime forecasting, while offering a distinct contribution in terms of interpretability and operational applicability. Existing studies have demonstrated that both statistical and machine learning models can effectively capture container throughput dynamics, particularly in multi-port or aggregated settings [34]. However, these approaches often focus primarily on predictive performance, with limited attention given to the interpretability of model outputs and their practical use in operational decision-making.

From a baseline perspective, traditional statistical models such as ARIMA, SARIMA, and Holt–Winters remain widely used due to their simplicity, stability, and transparency. These models provide reliable benchmark performance in many port forecasting applications, particularly when demand patterns are relatively stable. However, their ability to capture nonlinear relationships, structural breaks, and the influence of external operational variables is limited.

In contrast, machine learning and deep learning approaches, including ensemble models and LSTM-based architectures, have shown improved predictive accuracy in complex and high-dimensional settings. These models are capable of capturing nonlinear dependencies and temporal dynamics more effectively than classical approaches. However, they often operate as black-box systems, making it difficult to interpret their outputs. In addition, their

implementation typically requires substantial computational resources and technical expertise, which may limit their practical deployment in operational port environments.

The proposed Prophet–SHAP framework is positioned between these two approaches. It extends the capabilities of classical time-series models by incorporating changepoints, seasonal patterns, and external regressors, while maintaining a transparent and modular structure. At the same time, the integration of SHAP provides a systematic way to interpret model outputs at the feature level, enabling a clear understanding of how key variables influence forecasting results.

This positioning implies an explicit trade-off. The objective of the present study is not to achieve the highest possible predictive accuracy across all scenarios, but to provide a balanced solution that combines reasonable forecasting performance with interpretability, computational efficiency, and ease of implementation. In this sense, the proposed framework is designed as a practical decision-support tool rather than a purely performance-driven forecasting model.

Furthermore, the application of the framework to a single-port context represents an additional contribution to the literature. While many existing studies focus on aggregated or multi-port systems, the present work emphasizes localized dynamics and operational constraints, allowing for a more direct linkage between forecasting outputs and real-world decision-making processes.

Overall, this study contributes to the existing literature by demonstrating that forecasting models in port logistics can be both effective and interpretable. By explicitly addressing the trade-offs between accuracy, complexity, and transparency, the proposed approach offers a practical and scalable solution aligned with the needs of modern port operations.

5.4 Model Performance Under Trade Disruptions

The evaluation under simulated trade volatility conditions highlights both the strengths and limitations of the proposed framework. During disruption scenarios, the model experienced a decline in performance, with increased error metrics and reduced explanatory power. This behavior reflects the inherent difficulty of capturing abrupt external shocks using models trained on historical patterns.

However, the model demonstrated a strong recovery once system conditions stabilized, indicating its adaptability in dynamic environments. These findings are aligned with previous research emphasizing the sensitivity of forecasting models to exogenous factors not explicitly included in the model structure.

From a practical standpoint, this suggests that while Prophet provides a robust baseline forecasting tool, its performance can be further enhanced through the integration of real-time external indicators or anomaly detection mechanisms. Such enhancements would improve resilience under high-volatility conditions and support more responsive decision-making.

5.5 Practical and Managerial Implications

The proposed framework offers clear practical value for port operators and decision-makers. By combining accurate forecasting with interpretability, the model enables more informed and transparent planning across key operational areas.

Specifically, the results support:

- More efficient quay crane scheduling based on anticipated vessel arrivals;
- Improved AGV dispatching aligned with expected workload variations;
- Better yard space management during peak demand periods; and
- Enhanced truck appointment systems to mitigate congestion.

These applications demonstrate how forecasting outputs can be translated into actionable decisions, contributing to improved operational efficiency and reduced system bottlenecks.

More broadly, the study highlights the potential of explainable AI to support the digital transformation of port systems. By providing both predictive insights and explanatory transparency, the proposed approach aligns with the evolving requirements of smart and data-driven logistics environments [35, 36].

To further support the interpretation of the results, Table 3 summarizes the key findings and their corresponding operational implications in the context of port logistics.

The table demonstrates how the proposed framework translates predictive results into actionable insights, reinforcing its value as a decision-support tool in dynamic port environments.

The operational impact of forecasting errors becomes particularly evident under disruption scenarios. For example, an increase in forecasting error from approximately 300 TEU under stable conditions to over 600 TEU during disruption may lead to significant inefficiencies in resource allocation. Such deviations can result in either underutilization of quay cranes during overestimated demand or congestion and delays when throughput is underestimated.

Table 3. Interpretation of key findings and operational implications

Key Finding	Interpretation	Operational Implication
MAE = 3.84% (~4, 200 TEU)	High forecasting accuracy with low deviation from actual throughput	Reliable support for medium-term planning and resource allocation
Strong annual seasonality (Nov-Jan peaks)	Throughput is influenced by global trade cycles and seasonal demand patterns	Proactive planning of crane allocation, yard capacity, and workforce deployment
Vessel frequency (highest SHAP impact)	Primary driver of container volume fluctuations	Align quay crane scheduling with vessel arrival intensity
Trade volume index (positive influence)	Reflects macro-level demand trends affecting throughput	Use as an early indicator for adjusting forecasting and planning decisions
Holidays (mixed impact)	Context-dependent influence on port operations	Adjust staffing levels and operational schedules accordingly
Performance drop during disruption ($R^2 = 0.55$)	Model sensitivity to unexpected external shocks	Integrate external indicators or anomaly detection mechanisms to enhance robustness
Recovery phase improvement	Model performance stabilizes after disruption	Suitable for deployment in relatively stable operational environments

Note: MAE: mean absolute error; TEU: Twenty-foot Equivalent Unit; SHAP: SHapley Additive exPlanations; R^2 : Coefficient of Determination.

From a practical standpoint, these discrepancies may affect multiple operational layers, including yard congestion levels, AGV scheduling efficiency, and truck turnaround times. Even moderate forecasting errors at the monthly level can translate into substantial cumulative impacts on port performance, particularly during peak periods [37, 38].

These findings emphasize the importance of integrating forecasting models with adaptive operational strategies. By anticipating not only expected throughput but also potential error margins, port operators can implement buffer mechanisms and contingency plans to mitigate the effects of uncertainty. This strengthens the role of the proposed framework as a decision-support tool rather than a purely predictive system.

6 Limitations

While the proposed Prophet–SHAP framework demonstrates strong performance and practical relevance for container throughput forecasting, several limitations should be acknowledged to contextualize the findings and guide future research.

First, the study is based on a single-port case (Mwani, Qatar). Although this localized approach allows for a detailed and context-specific analysis, it may limit the generalizability of the results to other ports with different operational structures, traffic patterns, or data availability. Extending the framework to multi-port or cross-regional datasets would provide further validation and enhance its applicability.

Second, the set of external variables incorporated in the model is relatively limited. The analysis focuses on vessel frequency, holidays, trade volume index, and container handling hours. While these variables capture key operational drivers, other influential factors, such as weather conditions, labor disruptions, geopolitical events, and inland transportation constraints, were not included due to data availability. Incorporating a broader set of exogenous variables could improve model sensitivity to real-world dynamics.

Third, the model relies on a static input structure, assuming that the relationships between variables remain stable over time. In practice, these relationships may evolve due to macroeconomic shifts or changes in trade patterns. This highlights the need for periodic model recalibration or adaptive learning mechanisms to maintain forecasting accuracy in changing environments.

Fourth, the analysis is based on historical monthly data and does not incorporate real-time data streams. While this aligns with medium-term planning requirements, real-world deployment would benefit from integrating streaming data and dynamic updating mechanisms to support real-time decision-making.

Finally, a direct benchmarking against deep learning models such as LSTM and DeepAR is not included in this study. Although such models are known for their strong predictive capabilities, they often require extensive computational resources and lack inherent interpretability. The Prophet–SHAP framework was deliberately selected to prioritize transparency, ease of implementation, and operational usability. Future research will include systematic comparisons with deep learning approaches to assess trade-offs between accuracy, complexity, and explainability [39].

7 Conclusions

This study developed an interpretable forecasting framework for container throughput in port logistics by integrating the Facebook Prophet model with SHAP. Using real operational data from Mwani, Qatar, the proposed approach demonstrated its ability to generate accurate short- and medium-term forecasts while providing clear insights into the factors driving throughput dynamics.

The empirical results confirm that the model effectively captures long-term trends and annual seasonal patterns inherent in container port operations. The achieved forecasting accuracy, reflected in a low MAE, indicates that the model is well suited for supporting medium-term planning decisions. These findings are consistent with prior research highlighting the effectiveness of Prophet-based approaches in logistics forecasting contexts.

A key contribution of this study lies in the integration of explainability into the forecasting process. By applying SHAP, the model provides transparent interpretations of how external variables such as vessel frequency, trade volume index, and holidays influence predicted throughput. This addresses a critical limitation of many machine learning models, which often achieve high accuracy at the expense of interpretability. The ability to link predictive outputs with operational drivers enhances trust in the model and supports more informed decision-making.

From an operational perspective, the proposed framework offers practical value across multiple logistics functions. Forecast outputs can be used to support quay crane allocation, optimize AGV scheduling, improve yard space management, and enhance truck appointment systems. By translating predictive insights into actionable planning inputs, the model contributes to improving efficiency and reducing congestion in port environments. These findings align with the broader shift toward data-driven and intelligent port management systems.

The analysis of model performance under simulated trade volatility further highlights both the strengths and limitations of the proposed approach. While forecasting accuracy declines under disruption conditions, the model demonstrates the ability to recover once system stability is restored. This suggests that the framework is robust under normal operating conditions but can benefit from integration with real-time external indicators or anomaly detection mechanisms to improve resilience in highly dynamic environments.

Overall, this study contributes to the growing body of research on XAI in maritime logistics by providing a scalable and interpretable forecasting solution tailored to single-port systems. By aligning predictive accuracy with operational interpretability, the proposed framework supports the transition from reactive logistics management to proactive and data-driven decision-making. This approach offers a practical foundation for advancing smart port initiatives and enhancing the resilience and efficiency of modern port operations.

8 Future Directions

Future research can extend the proposed framework along several methodological and practical dimensions to enhance its robustness, scalability, and operational relevance.

First, the integration of the forecasting framework with Port Community Systems (PCSs) represents a key direction for real-world deployment. This can be achieved through real-time data streaming pipelines, such as API-based data ingestion, combined with edge-level forecasting modules. Such integration would enable dynamic, low-latency decision support and allow port operators to respond more effectively to rapidly changing operational conditions.

Second, the incorporation of additional exogenous variables could improve the model's sensitivity to external influences. Future studies may include macroeconomic indicators, weather conditions, geopolitical events, and inland transportation constraints to better capture the complexity of container throughput dynamics. This would enhance the model's ability to operate under highly volatile and uncertain environments.

Third, comparative evaluation with advanced deep learning models, such as LSTM networks and DeepAR, is necessary to assess performance trade-offs between predictive accuracy, computational complexity, and interpretability. Such benchmarking would provide a more comprehensive understanding of the advantages and limitations of the proposed Prophet–SHAP framework.

Fourth, extending the framework to multi-port or multimodal logistics systems would allow for the analysis of interdependencies between ports and transport networks. This would support more comprehensive forecasting and enable coordinated decision-making across interconnected logistics systems.

Finally, the integration of anomaly detection techniques and adaptive learning mechanisms could further improve the model's robustness under disruption scenarios. By identifying structural breaks and unusual patterns in real time, the forecasting system could dynamically adjust to changing conditions and maintain predictive performance during periods of instability.

Author Contributions

Conceptualization, A.C. and B.L.; methodology, A.C.; software, B.L.; validation, A.C. and B.L.; formal analysis, A.C.; resources, A.C.; data curation, B.L.; writing—original draft preparation, A.C. and B.L.; writing—review and

editing, A.C. and B.L.; visualization, B.L.; supervision, A.C.; project administration, A.C. and B.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data that support the findings of this study are not publicly available due to confidentiality restrictions imposed by Mwani Qatar. However, the modeling framework and analysis procedures can be made available by the authors upon reasonable request.

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Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] A. Alavi, H. O. Nguyen, J. Fei, and J. Sayareh, "Port logistics integration: Challenges and approaches," 2018. <https://hdl.handle.net/102.100.100/550054>
- [2] P. Yan, L. Zhang, Z. Feng, and J. Zhang, "Research on logistics demand forecast of port based on combined model," *J. Phys.: Conf. Ser.*, vol. 1168, no. 3, p. 032116, 2019. <https://doi.org/10.1088/1742-6596/1168/3/032116>
- [3] L. Bouzidi, "Comparative forecasting of container throughput for maritime logistics using statistical, machine learning and explainable AI models," *Marit. Bus. Rev.*, pp. 1–17, 2025. <https://doi.org/10.1108/MABR-08-2025-0086>
- [4] T. Caimao, Y. Wei, and Y. Jiantao, "Quay crane scheduling in automated container terminal for the trade-off between operation efficiency and energy consumption," *Adv. Eng. Inform.*, vol. 48, p. 101285, 2021. <https://doi.org/10.1016/j.aei.2021.101285>
- [5] S. Xu, S. Zou, J. Huang, W. Yang, and F. Zeng, "Comparison of different approaches of machine learning methods with conventional approaches on container throughput forecasting," *Appl. Sci.*, vol. 12, no. 19, p. 9730, 2022. <https://doi.org/10.3390/app12199730>
- [6] D. Octaviansyah, E. Buchari, J. Arliansyah, and N. Nawawi, "Multicriteria analysis as a method for selecting the best route of hinterland connections: Case study in South Sumatra, Indonesia," *Asian J. Shipp. Logist.*, vol. 40, no. 1, pp. 22–29, 2024. <https://doi.org/10.1016/j.ajsl.2023.12.002>
- [7] E. Žunić, K. Korjenić, K. Hodžić, and D. Đonko, "Application of Facebook's Prophet algorithm for successful sales forecasting based on real-world data," *Int. J. Comput. Sci. Inf. Technol.*, vol. 12, no. 2, pp. 23–36, 2020. <https://doi.org/10.5121/ijcsit.2020.12203>
- [8] G. Kim, E. Lee, and B. Kim, "Strategic port management by consolidating container terminals," *Asian J. Shipp. Logist.*, vol. 38, pp. 19–24, 2021. <https://doi.org/10.1016/j.ajsl.2021.11.003>
- [9] L. Liço, I. Enesi, and H. Jaiswal, "Predicting customer behavior using Prophet algorithm in a real time series dataset," *Eur. Sci. J.*, vol. 17, no. 25, p. 10, 2021. <https://doi.org/10.19044/esj.2021.v17n25p10>
- [10] M. Eskafi, M. Kowsari, A. Dastgheib, G. F. Ulfarsson, G. Stefansson, P. Taneja, and R. I. Thorarinsdottir, "A model for port throughput forecasting using Bayesian estimation," *Marit. Econ. Logist.*, vol. 23, pp. 348–368, 2021. <https://doi.org/10.1057/s41278-021-00190-x>
- [11] M. Khayyat, K. Laabidi, N. Almalki, and M. Al-Zahrani, "Time series Facebook Prophet model and Python for COVID-19 outbreak prediction," *Comput. Mater. Contin.*, vol. 67, no. 3, pp. 3781–3793, 2021. <https://doi.org/10.32604/cmc.2021.014918>
- [12] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice (2nd ed.)*. Melbourne, Australia: OTexts, 2018.
- [13] M. Eskafi, M. Kowsari, A. Dastgheib, G. F. Ulfarsson, G. Stefansson, P. Taneja, and R. I. Thorarinsdottir, "A model for port throughput forecasting using Bayesian estimation," *Marit. Econ. Logist.*, vol. 23, no. 2, pp. 348–368, 2021. <https://doi.org/10.1057/s41278-021-00190-x>
- [14] C. Mi, R. Xu, and C. T. Lin, "Real-time recognition of smartphone user behavior based on Prophet algorithms," *arXiv preprint*, 2019, Art. no. arXiv:1909.08997. <https://doi.org/10.48550/arXiv.1909.08997>
- [15] N. A. Michail and K. D. Melas, "Containership new-building orders and freight rate shocks: A "wait and see" perspective," *Asian J. Shipp. Logist.*, vol. 39, no. 1, pp. 30–37, 2023. <https://doi.org/10.1016/j.ajsl.2023.01.001>

- [16] N. R. Dayama, A. Ernst, M. Krishnamoorthy, V. Narayanan, and N. Rangaraj, "New models and algorithms for the container stack rearrangement problem by yard cranes in maritime ports," *EURO J. Transp. Logist.*, vol. 6, no. 4, pp. 307–348, 2017. <https://doi.org/10.1007/s13676-016-0098-8>
- [17] P. C. Awah, H. Nam, and S. Kim, "Short term forecast of container throughput: New variables application for the Port of Douala," *J. Mar. Sci. Eng.*, vol. 9, no. 7, p. 720, 2021. <https://doi.org/10.3390/jmse9070720>
- [18] P. L. Pallis, "Port risk management in container terminals," *Transp. Res. Procedia*, vol. 25, pp. 4411–4421, 2021. <https://doi.org/10.1016/j.trpro.2017.05.337>
- [19] S. Sun, Y. Zheng, Y. Dong, N. Li, Z. Jin, and Q. Yu, "Reducing external container trucks' turnaround time in ports: A data-driven approach under truck appointment systems," *Comput. Ind. Eng.*, vol. 174, p. 108787, 2022. <https://doi.org/10.1016/j.cie.2022.108787>
- [20] S. J. Taylor and B. Letham, "Forecasting at scale," *Am. Stat.*, vol. 72, no. 1, pp. 37–45, 2018. <https://doi.org/10.1080/00031305.2017.1380080>
- [21] P. C. Awah, H. Nam, and S. Kim, "Short term forecast of container throughput: New variables application for the port of Douala," *J. Mar. Sci. Eng.*, vol. 9, no. 7, p. 720, 2021. <https://doi.org/10.3390/jmse9070720>
- [22] N. Truong, B. Le Ngoc, and K. Hwan Seong, "Data analytics and throughput forecasting in port management systems against disruptions: A case study of Busan Port," *Marit. Econ. Logist.*, vol. 25, pp. 61–89, 2023. <https://doi.org/10.1057/s41278-022-00247-5>
- [23] Z. H. Munim, C. S. Fiskin, B. Nepal, and M. M. H. Chowdhury, "Forecasting container throughput of major Asian ports using the Prophet and hybrid time series models," *Asian J. Ship. Logist.*, vol. 39, no. 2, pp. 67–77, 2023. <https://doi.org/10.1016/j.ajsl.2023.02.004>
- [24] Z. Xing, H. Liu, T. Wang, E. P. Chew, L. H. Lee, and K. C. Tan, "Integrated automated guided vehicle dispatching and equipment scheduling with speed optimization," *Transp. Res. Part E: Logist. Transp. Rev.*, vol. 169, p. 102993, 2023. <https://doi.org/10.1016/j.tre.2022.102993>
- [25] A. Huang, X. Liu, C. Rao, Y. Zhang, and Y. He, "A new container throughput forecasting paradigm under COVID-19," *Sustainability*, vol. 14, no. 5, p. 2990, 2022. <https://doi.org/10.3390/su14052990>
- [26] Y. Özüpak and S. Mansurov, "A hybrid deep learning approach for multi-output short-term electricity demand forecasting," *Concurr. Comput. Pract. Exp.*, vol. 37, no. 25–26, p. e70356, 2025. <https://doi.org/10.1002/cpe.70356>
- [27] Y. Özüpak, "Real-time detection of photovoltaic module faults using a hybrid machine learning model," *Sol. Energy*, vol. 302, p. 114014, 2025. <https://doi.org/10.1016/j.solener.2025.114014>
- [28] T. Notteboom, A. Pallis, and J. P. Rodrigue, *Port Economics, Management and Policy*. London, United Kingdom: Routledge, 2022.
- [29] T. N. Cuong, L. N. B. Long, H. S. Kim, and S. S. You, "Structural analysis of factors for revitalizing Lena River logistics using ISM method," *Asian J. Shipp. Logist.*, vol. 39, no. 2, pp. 46–51, 2023. <https://doi.org/10.1016/j.ajsl.2023.02.001>
- [30] Z. H. Munim, C. S. Fiskin, B. Nepal, and M. M. H. Chowdhury, "Forecasting container throughput of major Asian ports using the Prophet and hybrid time series models," *Asian J. Shipp. Logist.*, vol. 39, no. 2, pp. 67–77, 2023. <https://doi.org/10.1016/j.ajsl.2023.02.004>
- [31] H. K. Chan, S. Xu, and X. Qi, "A comparison of time series methods for forecasting container throughput," *Int. J. Logist. Res. Appl.*, vol. 22, no. 3, pp. 294–303, 2019. <https://doi.org/10.1080/13675567.2018.1525342>
- [32] M. Niu, Y. Hu, S. Sun, and Y. Liu, "A novel hybrid decomposition-ensemble model based on VMD and HGWO for container throughput forecasting," *Appl. Math. Model.*, vol. 57, pp. 163–178, 2018. <https://doi.org/10.1016/j.apm.2018.01.014>
- [33] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," *arXiv preprint*, 2017, Art. no. arXiv:1705.07874. <https://doi.org/10.48550/arXiv.1705.07874>
- [34] Y. Feng, Y. Liu, and Y. Chen, "A robust multi-supplier multi-period inventory model with uncertain market demand and carbon emission constraint," *Comput. Ind. Eng.*, vol. 165, p. 107937, 2022. <https://doi.org/10.1016/j.cie.2022.107937>
- [35] V. Gosasang, W. Chandraprakaikul, and S. Kiattisin, "A comparison of traditional and neural networks forecasting techniques for container throughput at Bangkok port," *Asian J. Ship. Logist.*, vol. 27, no. 3, pp. 463–482, 2011. [https://doi.org/10.1016/S2092-5212\(11\)80022-2](https://doi.org/10.1016/S2092-5212(11)80022-2)
- [36] S. Shankar, P. V. Ilavarasan, S. Punia, and S. P. Singh, "Forecasting container throughput with long short-term memory networks," *Ind. Manag. Data Syst.*, vol. 120, no. 3, pp. 425–441, 2020. <https://doi.org/10.1108/IMDS-07-2019-0370>
- [37] S. Xu, S. Zou, J. Huang, W. Yang, and F. Zeng, "Comparison of different approaches of machine learning methods with conventional approaches on container throughput forecasting," *Appl. Sci.*, vol. 12, no. 19, p.

9730, 2022. <https://doi.org/10.3390/app12199730>

- [38] S. Tang, S. Xu, and J. Gao, “An optimal model based on multifactors for container throughput forecasting,” *KSCE J. Civ. Eng.*, vol. 23, no. 9, pp. 4124–4131, 2019. <https://doi.org/10.1007/s12205-019-2446-3>
- [39] Y. Özüpak and S. Mansurov, “Optimizing electricity demand forecasting with a novel RNN-LSTM hybrid model,” *Energy Sources, Part B: Econ. Planning, Policy*, vol. 20, no. 1, p. 2531448, 2025. <https://doi.org/10.1080/15567249.2025.2531448>